

LEM16_18 Definition

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2) **LEM:** Logic Emulation Machine (variable bit size fields in memory)

2.a) Characterization:

Single or dual accumulators
Index (word mode) and size registers (bit-field mode: size, bit delta, word address)
Lookup Table instructions (some part of accumulator is index in to memory LUT)
Dual issue by including orthogonal set of word instructions
Goals: small LUT count (<400), high frequency of operation, 1 & 2+ pipeline stages

2.b) Instance: LEM16_18:

16-bit data, 18-bit instructions
Bit field instructions to/from accumulator
Word instructions to/from accumulator
Bit field instructions operate to/from data RAM
Word instructions operate to/from program RAM
Single port RAM supports return stack, accumulator stack and index registers
Bit fields do not cross word boundaries.

128	000IIISSSMMMMMMMM	Bit field inst using size constant
128	001IIISSSMMMMMMMM	Bit field inst using size constant
128	010IIISSSMMMMMMMM	Bit field inst using size constant
32	011III00XXMMMMMMMM	Bit field inst using index register
32	011III01XXMMMMMMMM	Bit field inst using index register
32	011III10XXMMMMMMMM	Bit field inst using index register
ADD, ADC, SUB, SBC, RADD, RADC, RSUB, RSBC, LD, AND, OR, XOR, ST, RAND, ROR, RXOR, LDC, BIT/TST, CMP, STC, CLR, SET, INC, DEC		
32?	011III11IUMMMMMMMMM	Immediat inst, U selects upper/lower use of immediate
ADDI, ADDIU, SUBI, SUBIU, BITI, BITIU, CMPI, CMPIU, LDI, LDIU, ANDI, ANDIU, ORI, ORIU, XORI, XORIU, ADCI, SBCI, LDNI, LDPI, ADXnI, LDXnI, LDXnIU		
128	100YIIIIIXXMMMMMMMM	Word inst, Y selects index & 8-bit M field versus 10-bit M field
ADD, ADC, SUB, SBC, AND, ANDC, OR, XOR, LD, LDC, BIT, CMP, ST, STC, CLR, SET		
128	110YCCCCXXMMMMMMMM	JMP, Y selects index & 8-bit M field versus 10-bit M field
128	111YCCCCXXMMMMMMMM	CALL, Y selects index & 8-bit M field versus 10-bit M field
32?	101YCCCCXXMMMMMMMM	Word2 inst, Y selects index & 8-bit M field versus 10-bit M field
LUT, MUL, MULU, MULS		
16	1011CCCC00MMMMMMMM	Bcc
16	1011CCCC01MMMMMMMM	RTNIcc
16	1011CCCC10MMMMMMMM	PRED
16	1011IIIIU11MMMMMMMM	MISC
ORISR, ANISR, MULI, MULIU, MULUI, MULUIU, MULSI, MULSIU, BREV, BRODD, BRNA1S, BRA1S, BRPARE, BRPAROD, LDXSZI, Others		
1	1011111111IIIIIMMM	Others
MA2X, MX2A, RTNCC, ROTATE, Others2		
1/16	10111111110100IIII	Others2
MA2X, MSR2A, POPCT, LDZCT, TRZCT, HALT, WAIT, TRAP, PUSHA, PUSHPC, POPA, DISCARD, LOOP, CLWD		
SSSSNNNNNNNNNNNNNN		Four Bit field index registers
NNNNNNNNNNNNNNNNNN		Four program memory index registers or Eight return addresses
NNNNNNNNNNNNNNNNNN		16-bit Accumulator, 10+ bit Program counter
NNNNNNNNNNNNNNNNNN		?-bit Status register
CCCCDDDEEEEE		12-bit Predicate register
ADXnI, LDXnI, LDXnIU: M is 6-bits, n is 3-bits (U&M); LDXSZI: M is 4-bits, n is 2-bits (M)		

2.c) Instance: LEM16_18M: Minimal LEM16_18

16-bit data, 18-bit instructions

Bit field instructions to/from single accumulator

Bit field inst operations to/from data RAM (16 words = 256 bits)

To RAM: ST, SET, CLR, INC, DEC only

Bit fields do not cross word boundaries.

Single port RAM supports return stack, accumulator stack and index registers

Shared Return & Data stack (16x16, 8 locations each)

No index registers and minimal RAM & ROM (16x16, 1Kx18)

For Spartan-6: use LUT based shifter instead of Multiplier

For comparisons use 2-stage pipe version

Spartan-3, Spartan-6, Virtex-6, Stratix III, Cyclone II & Lattice ECP

256	SSSS00IIIIIMMMMMMM	Bit field inst using size constant
ADD, ADC, SUB, SBC, LD, AND, OR, XOR, LDC, TST, CMP, ST, CLR, SET, INC, DEC		
256	CCCC01UIIIIMMMMMMM	Immediate instruction, U selects upper/lower use of immediate
ADDI, ADDIU, SUBI, SUBIU, BITI, BITIU, CMPI, CMPIU, LDI, LDIU, ANDI, ANDIU, ORI, ORIU, XORI, XORIU		
64	CCCC1000MMMMMMMMMM	JMP
64	CCCC1001MMMMMMMMMM	CALL
20	IIII1010MMMMMMMMMM	LUT, trailing comma code in address field sets table size
LUT1, LUT2, LUT4, LUT8, LUT16		
16	IIII1010MMMMMMMMMM	LUT, table size is single word
LUT1B, LUT2B, LUT4B, LUT8B		
12	IIII1011IIIMMMMMMM	LUT immediate (should these have a CC field?)
LUT1I, LUT2I, LUT4I		
16	CCCC101111DDDDDEEEE	1, 2 or 4-bit result
PRED (optional, CC field is predicate for next inst)		
196	CCCC11IIIIIMMMMMMM	MISC
ORISR, ANISR, ADCI, SBCI, RTN, RTNI		
HALT, WAIT, TRAP, PUSHA, PUSHPC, POPA, DISCARD, NOP		
NNNNNNNNNN		
10-bit Program counter		
NNNNNNNNNN		
Return address entries		
NNNNNNNNNNNNNNNNNN		
16-bit Accumulator		
CZNV A O P I		
8-bit Status register		
For truly minimal implementation make all CC except C a function of Accumulator		
CCCCDDDDDEEEE		
12-bit Predicate register (optional)		

SSSS00IIIIIMMMMMMM and IIII101IIIIIMMMMMMM do not have CC field

2.d) Instance: LEM16_18W: Word only LEM16_18 (e.g. bit sizes always 16)

128	000IIII0000MMMMMMMM	Bit field inst using 16-bit size constant
128	001IIII0000MMMMMMMM	Bit field inst using 16-bit size constant
128	010IIII0000MMMMMMMM	Bit field inst using 16-bit size constant
ADD, ADC, SUB, SBC, LD, AND, OR, XOR, ST, LDC, BIT/TST, CMP		
32?	011IIII11IUIMMMMMMM	Immediate inst, U selects upper/lower use of immediate
ADDI, ADDIU, SUBI, SUBIU, BITI, BITIU, CMPI, CMPIU, LDI, LDIU, ANDI, ANDIU, ORI, ORIU, XORI, XORIU, ADCI, SBCI, LDNI, LDPI, ADXnI, LDXnI, LDXnIU		
128	110YCCCCXXMMMMMMMM	JMP, Y selects index & 8-bit M field versus 10-bit M field
128	111YCCCCXXMMMMMMMM	CALL, Y selects index & 8-bit M field versus 10-bit M field
32?	101YCCCCXXMMMMMMMM	Word2 inst, Y selects index & 8-bit M field versus 10-bit M field
MUL, MULU, MULS		
16	1011CCCC00MMMMMMMM	Bcc
16	1011CCCC01MMMMMMMM	RTNIcc

16	10111111U11MMMMMMMM	MISC
	ORISR, ANISR, MULI, MULIU, MULUI, MULUIU, MULSI, MULSIU, BREV, BRODD, BRNA1S, BRA1S, BRPARE, BRPAROD, LDXSZI, Others	
1/16	101111111101001111	Others2
	MA2X, MSR2A, HALT, WAIT, TRAP, PUSHA, PUSHPC, POPA, DISCARD	
	SSSSNNNNNNNNNNNNNN	Four Bit field index registers
	NNNNNNNNNNNNNNNNNN	Four program memory index registers or Eight return addresses
	NNNNNNNNNNNNNNNNNN	16-bit Accumulator, 10+ bit Program counter
	CZNVAOPI	8-bit Status register
	ADXnI, LDXnI, LDXnIU: M is 6-bits, n is 3-bits (U&M); LDXSZI: M is 4-bits, n is 2-bits (M)	